

Deep Learning Algorithm for Atomization Characterization using Shadowgraph Images

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Abstract—Predicting droplet size distribution as a function of chemical composition, input pressure, nozzle geometry, and atmospheric conditions using algebraic approach is extremely complicated due to the number of variables involved, especially for agricultural spray applications. A low cost shadowgraphy technique was used to capture the spray pattern from six different standard ASABE nozzles (ranging from extremely coarse 6515 type to very fine 11001 nozzle type) at different inlet pressures, along with various combinations of adjuvant mixtures. The resultant images were then fed into convolution neural network (CNN) architecture to classify the different nozzle types, pressure, and the adjuvant mixtures. Preliminary results show that the CNN architecture was able to classify the spray patterns with different nozzles, pressures, and adjuvant mixtures with more than 98% accuracy. The architecture was able to identify the subtle changes in the atomization in different configurations which is shown through class activation mapping.

Index Terms—Agricultural spray, Shadowgraph, Spray pattern, Deep learning, Machine Learning

I. INTRODUCTION

Spray drift is one of the major concerns in agricultural industry which affects nozzle manufacturers, chemical manufacturers and more importantly, farmers who are the applicators of the chemicals. Over the course of many years, spray drift issues have led to loss of lives and several law suits between the three dependent branches – farmers, nozzle manufacturers and chemical manufacturers [1]. Three-fourths of agriculture-related complaints investigated by the Ohio Department of Agriculture involved drift issues. Two-thirds of the total complaints in a five-year period brought to the attention of Iowa Department of Agriculture were related to drift problems and one-third of court cases due to spray misapplications reported by a major insurance company involved drift damages [2]. With the advent of genetically modified crops, the use of non-selective herbicides will be restricted because even a small amount of these herbicides can cause serious damage to off-target plantations. For example, applications of 2,4-D and Dicamba at levels as low as 1/300th of the field rate for soybeans caused statistically significant field loss in tomato crops. Drift of either herbicide caused a 17-77% reduction in the marketable fruit when applied a 1/100th of the field rate [3]. The number of herbicide applications is increasing substantially each year, with the U.S. alone increasing herbicide use by 500% between 2000 and 2014

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[4]. In 2017, the use of Dicamba on Monsanto's Dicamba-resistant crops made headlines when the uncontrolled drift affected millions of acres of farms from Arkansas to North Carolina [5]. With the increasing number of spray applications each year, the likelihood of off-target movement of the sprays is of great concern.

Despite the increased use and risk, it is believed that one of the main challenges has to do with how the chemicals are applied with a nozzle type by farmers. For instance, it is customary for farmers to mix chemicals (pesticides, adjuvants, etc.) during a spray application that has not been tested in a wind tunnel for spray drift potential in order to save time and effort. As such, the atomization of the fluid changes for the rated nozzle type and pressure in a manner that inadvertently increases the drift potential under certain conditions resulting in spray drift and consequential harm. Therefore, the goal of this research is to develop technology that can accurately classify any given agricultural spray into 6 categories: extremely coarse, very coarse, coarse, medium, fine, and very fine. The droplet size distributions for these different nozzles are shown in Table I.

TABLE I
DROPLET DIAMETER DISTRIBUTIONS FROM DIFFERENT ASABE NOZZLES

Nozzle Types	Droplet Distributions (μ %m)		
	Dv(10)	Dv(50)	Dv(90)
11001 (Very Fine)	64.7	141.8	245.7
11003 (Fine)	110.9	247.6	412.0
11006 (Medium)	161.2	352.0	561.2
8008 (Coarse)	201.3	442.3	739.3
6510 (Very Coarse)	243.1	528.6	854.2
6515 (Extremely Coarse)	337.1	693.6	1088.5

Droplets less than 100 μ m are especially prone to spray drift. The measurements in Table 1 are determined by the use of Spraytech Laser Diffraction instrument in the University of Dayton Low Speed Wind Tunnel. However, these types of measurements are extremely difficult to obtain in real-time during the actual spray application by the farmers. Therefore, advanced machine learning algorithms were trained by high speed shadowgraphy images of the spray pattern from different nozzles as well as images obtained by a traditional cell phone camera to classify a given spray pattern into the above six categories which will help farmers predict the spray drift potential of any given nozzle-chemical-pressure

combinations. Since the atomization of the droplets changes when different chemicals are mixed, standard spray drift prediction models such as AgDisp and AgDrift does not work. Moreover, chemical and nozzle companies introduce several new products every year, especially shear thickening adjuvants/chemicals in an effort to minimize spray drift which makes the existing prediction models obsolete [6]. Hence, a new method to characterize and predict the spray drift is needed for chemical mixtures which can adapt to new nozzle design and new viscoelastic chemical compositions. One primary advantage of machine learning algorithms is continuously assessing and learning from spray atomization images and recognizing features/patterns (such as fluid ligament formation, break up, droplet sizes, etc.) that are essential for identification of the chemical composition and nozzle type. In addition, the class activation mapping results [7] for our deep learning algorithms were studied in order to provide further insights about the prediction to the user.

A. Spray Atomization

An example shadowgraph image of a spray from a flat fan agricultural nozzle is shown in Figure 1. In this type of nozzle, the aerodynamics induces unstable short-wavelength waves that grows on the liquid sheet and breaks it up into droplets smaller than the nozzle orifice. Fraser et al. [8] proposed a breakup mechanism for these types of nozzles as shown in Fig. 1.

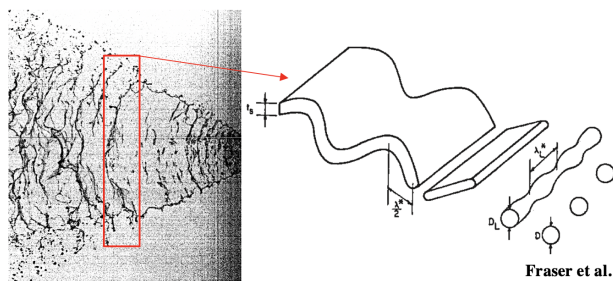


Fig. 1. Shadowgraph Image of a Spray and the Breakup Mechanism proposed by Fraser et al.[8]

In this mechanism, a sinuous Kelvin-Helmholtz instability grows until the sheet necks down to a critical thickness value, at which point surface tension forces cause the sheet to rupture into spanwise ligaments, which assume a roughly cylindrical shape. These cylindrical ligaments then break down via the Rayleigh mechanism into spherical droplets. This entire process is known as atomization, which is highly dependent on the properties of the fluid and the nozzle. Most characterization of atomization process has only been conducted on Newtonian fluids, such as water and oil – relatively thin, homogeneous liquids that don't include any long polymers. It is customary for agricultural applications to use chemical agents (fluid thickening agents/adjuvants) to reduce spray drift which changes the viscoelastic

properties of the fluid, which in turn highly influences the atomization process. Dambrowski et al. [9] showed that the in-homogeneity in sprays involving oil-water emulsion types leads to perforations (hole formation) on the liquid sheet prior to ligament formation which increases the characteristic thickness of the liquid sheet upon breakup into ligaments, presumably generating coarser spray droplets. The atomization process leads to a droplet size distribution which can influence the rate of on-target deposition as well as canopy penetration, with smaller droplets providing better deposition and canopy penetration than larger ones [9]. However, droplet sizes below 100 μ m in diameter are the most prone to spray drift [10], [11]. Along with the droplet size, droplet velocity also affects the structure of the spray deposits and the “driftability” of the droplets [12]. Therefore, it is imperative that a robust prediction model be developed which 1) Provides reasonable accurate predictions of atomization process occurring from a given chemical composition and a nozzle type. 2) Is adaptable to the new nozzle designs and chemical compositions which are manufactured by nozzle and chemical companies.

II. METHODOLOGY

All experiments were conducted in the spray test chamber at the University of Dayton Low Speed Wind Tunnel laboratory. The spray test chamber houses the nozzle on a vertical traverse, a constant pressure pump, and several stage filters to filter any chemicals that are sprayed in the chamber. The two windows on either side of the test chamber allows to place the laser diffraction instrument as well as setup the necessary equipment for shadowgraphy technique. The maximum wind speed range with the spray drift chamber in the wind tunnel is 5 to 20 mph. The experiments were conducted at a freestream velocity of 15 mph. A 40W 1100 LM LED panel was used as a light source for the shadowgraphy technique along with a PCO 1200 camera for various chemical compositions, nozzle types and pressures. The chemical used in the current investigation was the InterLock adjuvant mixed with water in three mixture configurations: 0.5%, 1%, and 1.5%.

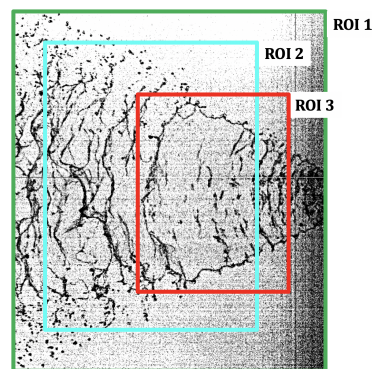


Fig. 2. Three different ROIs of Spray Pattern

III. IMAGE PRE-PROCESSING

The shadowgraph images and the different ROIs are shown in Fig. 2. We preprocess the dataset and solely provide the cropped ROI as input to the ML algorithm. The dataset is split into three subsets: 72% for training, 8% for validation, and 20% for testing.

IV. AUTOMATED ANALYSIS ARCHITECTURE

In this section, we present the machine learning architectures to identify chemical composition and nozzle respectively. Fig. 3 presents the top-level block diagram for automated analysis. Before we apply any deep learning architectures, we preprocess the images by increasing the contrast and cropping the region of interest. Table II presents the distribution of shadowgraph images collected for our experiment across different classes.

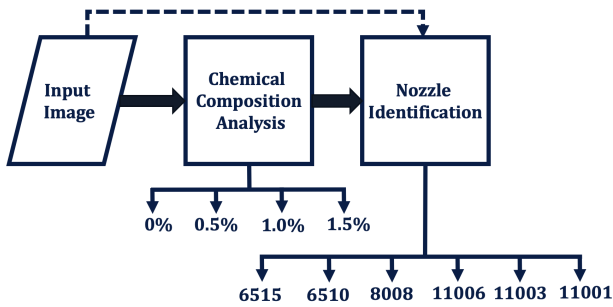


Fig. 3. Top-Level Block Diagram for the Analysis

TABLE II
OVERALL DATASET (NUMBER OF IMAGES) DISTRIBUTION

Nozzle Types	Chemical Compositions			
	Water (0%)	0.5%	1.0%	1.5%
11001 (Very Fine)	1396	51	1861	1857
11003 (Fine)	2583	1678	2781	2745
11006 (Medium)	2121	1945	2791	2906
8008 (Coarse)	2543	1623	2977	3326
6510 (Very Coarse)	2417	2325	2497	2978
6515 (Extremely Coarse)	3236	2197	2890	3233

V. RESULTS

A. Chemical Composition

In this section, we present the architecture and results obtained for chemical composition detection. We propose a deep learning architecture to identify the chemical composition from a given shadowgraph image. Note that we utilize images from all nozzles for chemical composition detection. We utilize 72% of the dataset for training, 8% for validation, and 20% for testing respectively. Due to limited training data, we adopt transfer-learning approaches utilizing an established network ResNet (ResNet-50) for classification purposes. We spatially re-sample all images to 224 x 224.

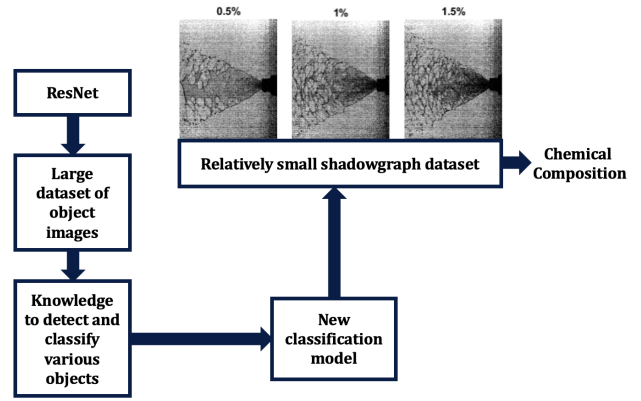


Fig. 4. Top-Level Block Diagram for the Transfer Learning Approach

Fig. 4 presents the top-level block diagram of the transfer learning approach implemented in this study. These approaches have proven to be highly effective with limited training images. Moreover, the study of such well-established networks would help us understand the effectiveness of such approaches for shadowgraph images. We study the performance of the algorithms using confusion matrices. Fig. 5 presents the confusion matrix obtained which indicates a 98.4% accuracy in chemical composition classification.

Confusion Matrix: ResNet

Output Class	0.5%	1%	1.5%	Water	
	0.5%	1%	1.5%	Water	
0.5%	1892 16.6%	3 0.0%	0 0.0%	2 0.0%	99.7% 0.3%
1%	69 0.6%	3097 27.2%	21 0.2%	40 0.4%	96.0% 4.0%
1.5%	3 0.0%	45 0.4%	3388 29.8%	0 0.0%	98.6% 1.4%
Water	0 0.0%	0 0.0%	0 0.0%	2817 24.8%	100% 0.0%
	96.3% 3.7%	98.5% 1.5%	99.4% 0.6%	98.5% 1.5%	98.4% 1.6%

Fig. 5. Confusion matrix obtained for chemical composition

The mean Class Activation Mapping (CAM) for different chemical compositions results are shown in Fig. 6 for the Medium (11006) nozzle. CAM is a visualization tool where the probabilities predicted by the network are mapped back to the final convolutional layer to highlight the discriminative regions that are specific to that class. It is the result of the activation map of the ReLU (Rectified Linear Unit) after the last convolutional layer for a particular class which is determined by how much each activation contributes to the

final probability score of that particular class. For an image, CAM represents the global average pooling layer applied following the last convolutional layer based on the spatial location in order to generate the weights. Therefore, it allows distinguishing the areas within an image that differentiates the class, which leads to its probability predictions. [13] In theory, an increase in the adjuvant concentration shifts the droplet diameter distributions to higher droplet diameter. It also changes the breakup process. As mentioned in the literature review section, the addition of adjuvant creates perforations in the liquid sheet which leads to thicker ligaments and a higher droplet diameter. The CAM results for the Water and 0.5% relies on the ligament formation zone for classification. However, the CAM results for the 1.0% and 1.5% mixture cases shows that the algorithm relies on the droplet formation zones for the classifications since the ligament formation zones for these two cases may be similar. The mean CAM results therefore suggests that the classification of the spray pattern into different chemical composition mixture maybe based on the physics of atomization.

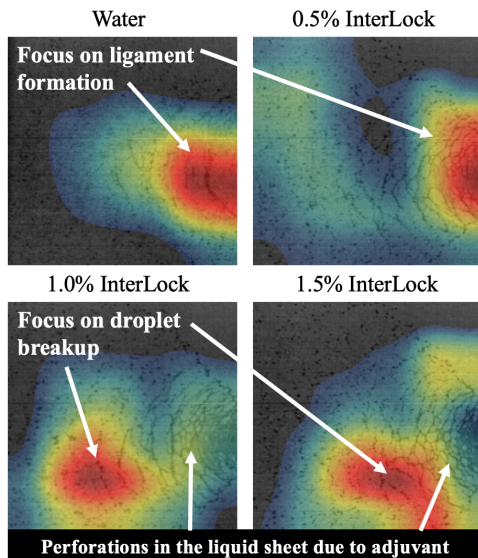


Fig. 6. Mean CAM results for chemical composition

B. Nozzle Identification

In this section, we present the architecture and results obtained for nozzle identification across each chemical composition. We adopt the same transfer learning approach using ResNet for identifying nozzles in each chemical composition i.e. for each chemical composition, an algorithm is trained to identify the nozzle type. Similar to chemical composition analysis, we study the performance by splitting the datasets into groups of 72%, 8%, and 20% for training, validation, and testing respectively.

The confusion matrix for the water, 0.5%, 1.0%, and 1.5% chemical compositions are shown in Fig. 7. The results from confusion matrix indicates that the accuracy of classifying

the different nozzles in the case of water is 100%, in the case of 0.5% chemical composition is 100%, in the case of 1.0% chemical composition is 99.9%, and in the case of 1.5% chemical composition is 100%. The slight reduction in the accuracy is due to higher similarities between the Extremely Coarse (6515) and Very Coarse (6510) nozzle with increasing chemical composition.

Similar to the chemical composition results, the mean class activation mapping (CAM) results were investigated for the nozzle identification classification. This is to ensure that the classification is performed based on the spray physics and atomization and not due to region of interest, edges, background noise, or any other factors unrelated to spray dynamics. The mean CAM results for the nozzle identification for the baseline water case is shown on Fig. 8

The mean CAM for the 6515 nozzle case is focused on the Kelvin-Helmholtz instability which is a precursor to the formation of ligaments that leads to atomization. The 6510 case however is only focused on the edge of the spray and perhaps the angle of the spray. This off-spray focus is hypothesized due to the similarities in between the 6515 and the 6510 nozzle. The 8008 case however is classified based on the ligament formation, right before the spray atomizes. Both the 11006 and 11003 nozzles are classified based on the spray droplet formation and dynamics soon after the ligament formation occurs. It can clearly be seen that the CAM is only focused on the spray and not on the ligaments for these two nozzles. Since majority of the field of view consists of droplets in the case of 11001 nozzle, the classification of this nozzle type is based on the droplet dynamics and trajectory.

Similar analysis can be performed on the nozzle identification for the 0.5% case as shown in Fig. 9. The addition of the adjuvant creates perforations in the liquid sheet before the ligament formation and the 6515 nozzle is classified based on these perforations as seen in the mean CAM results. This dynamics is very similar to the dynamics seen in 6510 nozzle, which is why it is hypothesized that the algorithm focused on the expansion of the spray instead of the main spray itself. However, with the 8008 nozzle case, the perforations leads to ligament formation which becomes the main focus for classifying the 8008 nozzles. The 11006 and 11003 nozzles are once again classified based on the droplet formations soon after the ligament breakup. And the 11001 nozzle is classified based on the droplets dynamics and trajectory since majority of the field of view is covered by the droplets.

The mean CAM results for the 1.0% chemical composition nozzle identification also show similar trends seen in the 0.5% case. The mean CAM results for the 1.0% chemical composition case is shown in Fig. 10. With increase in the concentration of the adjuvant, the presence of perforations is clearly seen in the 6515 case. The ligament formation is highly delayed by the addition of more adjuvant that the entirety of the spray in the field of view is only comprised of perforations which becomes the focus behind the 6515 nozzle classification. Once again, due to the similarities between the 6515 and the 6510 nozzle, the focus of the 6510 nozzle for classification

Confusion Matrix: ResNet							
Output Class	Nozzle 11001	Nozzle 11003	Nozzle 11006	Nozzle 6510	Nozzle 6515	Nozzle 8008	
Water	279 9.8%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	517 18.1%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	424 14.8%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	483 16.9%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	0 0.0%	647 22.6%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	599 17.8%	100% 0.0%
	100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%	
0.5% InterLock	10 0.5%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	336 17.1%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	389 19.8%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	465 23.7%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	0 0.0%	439 22.4%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	325 16.5%	100% 0.0%
	100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%	
1.0% InterLock	372 11.8%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	556 17.7%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	544 17.3%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	497 15.8%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	1 0.0%	578 18.4%	0 0.0%	99.8% 0.2%
	0 0.0%	0 0.0%	0 0.0%	1 0.0%	0 0.0%	595 18.9%	99.8% 0.2%
	100% 0.0%	100% 0.0%	100% 0.0%	99.6% 0.4%	100% 0.0%	100% 0.0%	99.9% 0.1%
1.5% InterLock	371 10.9%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	549 16.1%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	581 17.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	595 17.5%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	1 0.0%	647 19.0%	0 0.0%	99.8% 0.2%
	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	665 19.5%	100% 0.0%
	100% 0.0%	100% 0.0%	100% 0.0%	99.8% 0.2%	100% 0.0%	100% 0.0%	100.0% 0.0%
Target Class	Nozzle 11001	Nozzle 11003	Nozzle 11006	Nozzle 6510	Nozzle 6515	Nozzle 8008	

Fig. 7. Confusion Matrix for Nozzle Identification for all Chemical Compositions

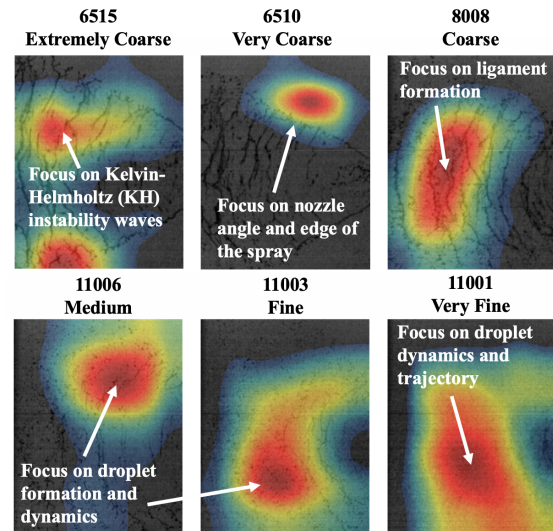


Fig. 8. Mean CAM for Nozzle Identification for Water

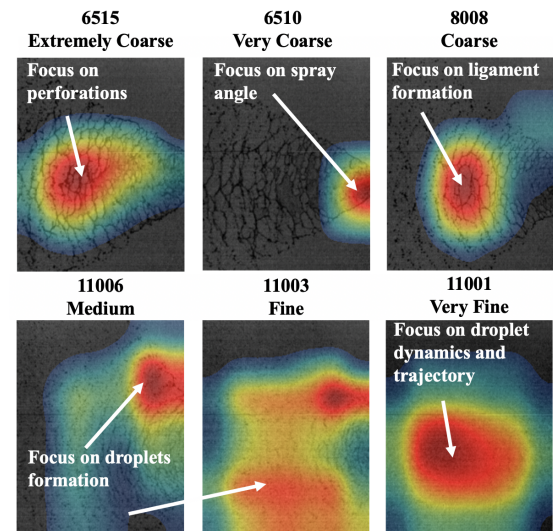


Fig. 9. Mean CAM for Nozzle Identification for 0.5% Chemical Composition

is at the beginning of the spray as seen in 0.5% composition case as well.

The focus of classification for the 8008 remains to be the ligament formation, however, part of the focus is also on the edge of the spray. Unlike in the above two cases, the mean CAM results for the 11006 and the 11003 nozzles are quite different. The 11006 case is classified based on the ligament and droplet formation near the nozzle. However, the 11003 case is classified based on the droplet formation and trajectory however, the higher weight is seen near the droplet formation zone. The mean CAM results for the 11003 and 11001 is a bit more similar than the previous two cases. However, in the case of 11001 nozzle, more weight is seen in the center of the field of view which captures the droplets dynamics and the

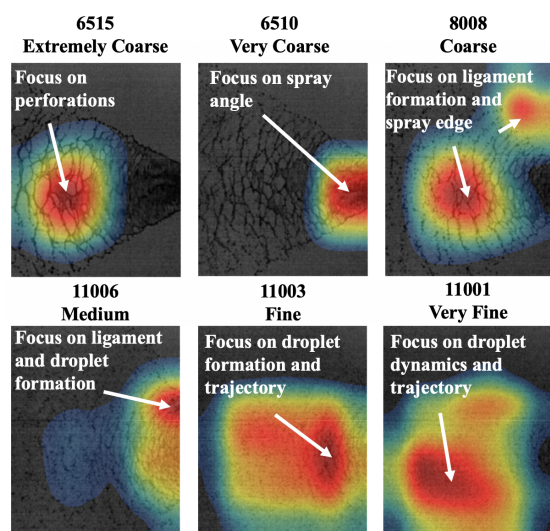


Fig. 10. Mean CAM for Nozzle Identification for 1.0% Chemical Composition

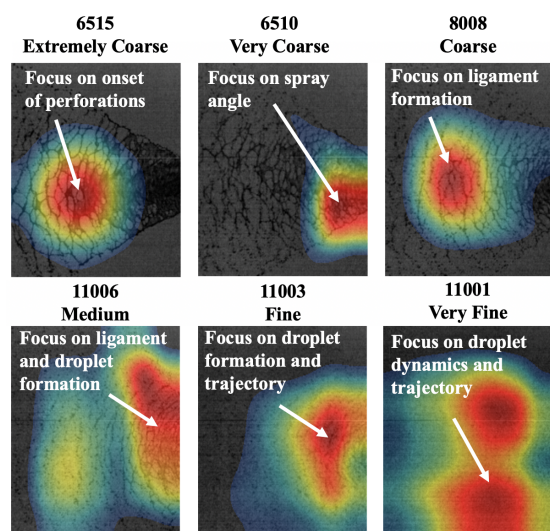


Fig. 11. Mean CAM for Nozzle Identification for 1.5% Chemical Composition

trajectory.

The nozzle identification for the 1.5% chemical composition is also investigated through the mean CAM results as shown in Fig. 11. With a higher adjuvant mixture, the ligament formation is pushed even further downstream in the case of 6515 nozzle that the majority of the field of view is covered by the perforations in the liquid sheet. Therefore, these perforations becomes the focus of identification for the 6515 nozzle. The focus of classification for the 6510 nozzle remains to be at the beginning of the spray, however covering the beginning parts of the instability process. The ligament formation remains as the focus of classification for the 8008 nozzle. The ligament and droplet formation towards the beginning of the spray remains as a marker for the 11006 nozzle whereas the droplet formation, dynamics, and trajectory remains to be the marker for the 11003 and 11001 nozzle.

VI. CONCLUSIONS

A low cost shadowgraphy technique was used to obtain high frame rate images of the spray with multiple chemical compositions emanating from six different standard ASABE nozzles with different input pressures and nozzle geometries. These images were then used to train a deep learning algorithm to classify a spray into six different categories: Extremely Coarse, Very Coarse, Coarse, Medium, Fine, and Very Fine. at four different chemical compositions generated by mixing InterLock adjuvant at different mixing ratios: 0% (Water), 0.5%, 1.0%, and 1.5% chemical compositions. The chemical composition classification algorithm was based on the transfer learning approach and gave an accuracy of 98.4%. Once the chemical compositions were identified, the nozzle identification algorithm trained by the transfer learning approach using ResNet was utilized to classify the six different nozzle types. The average classification accuracy for the nozzles were 99.9%. The mean CAM results were investigated to determine

the basis for classification for both the chemical and nozzle identification. The CAM results indicated that the classification was based on the spray physics such as ligament formation, perforations, droplet formation, dynamics, and trajectory.

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